National Skill Training Institute

MODULE
Artificial Intelligence & Machine Vision

TOPIC TITLE
Rice Variety Classification Using Deep Learning

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1. INTRODUCTION

The accurate classification of rice varieties is a significant challenge in agricultural science, with far-reaching implications for food quality control,supply chain management, and automated sorting systems. Rice is one of the most consumed staple foods globally, and distinguishing between varieties - such as Arborio, Basmati, Ipsala, Jasmine, and Karacadag - requires recognizing subtle morphological features that are difficult to differentiate by the naked eye at industrial scale.

The emergence of deep learning, particularly Convolutional Neural Networks (CNNs), has transformed image-based classification tasks. Unlike traditional machine learning approaches that rely on handcrafted feature extraction, CNNs are capable of learning hierarchical feature representations directly from raw pixel data. More recently, transfer learning has further accelerated progress by leveraging pre-trained models on large-scale datasets such as ImageNet, enabling high-accuracy classification even when training data is limited.

This project implements and compares two state-of-the-art transfer learning architectures - MobileNetV2 and InceptionV3 — for the classification of five rice grain varieties. The study evaluates each model on a balanced dataset of 3,000 images, reporting metrics including accuracy, precision, recall, F1-score, and ROC-AUC curves. The implementation was performed in Python using TensorFlow and Keras, executed on GoogleColab.

The motivation for this work extends beyond academic evaluation. Automated rice variety identification has practical applications in:

- Agricultural sorting and grading systems
- Food quality and authenticity verification
- Supply chain transparency and labelling compliance
- Research into grain morphology and crop genetics

This report details the complete pipeline from dataset preparation to final evaluation, providing a thorough comparative analysis of MobileNetV2 and InceptionV3 in a balanced, augmented training environment.

2. INNOVATIVE APPROACH

The core innovation of this project lies in the application of transfer learning to a domain-specific agricultural classification task. Rather than training a CNN from scratch - which would require substantially more data and compute - both MobileNetV2 and InceptionV3 were initialised with ImageNet pre-trained weights. This enables the models to leverage general visual representations learned from millions of images, which are then fine-tuned for rice grain classification.

MobileNetV2 employs depthwise separable convolutions and an inverted residual block structure, making it computationally lightweight while preserving representational capacity. This architecture is particularly suited to deployment on resource-constrained hardware, making it attractive for real-time agricultural applications.

InceptionV3 introduces a more complex multi-scale feature extraction approach using factorised convolutions and inception modules that process feature maps at multiple receptive field sizes simultaneously. This allows the model to capture both fine-grained texture differences between rice varieties and broader shape characteristics.

A key methodological contribution of this project is the application of comprehensive data augmentation during training. Rather than relying on raw images, the training pipeline applies random rotations (up to 25 degrees), zoom, width/height shifts, shear transformations, and horizontal flipping. These augmentations significantly increase effective dataset diversity, helping the models generalise beyond the specific imaging conditions of the source dataset.

Additionally, both models incorporate a custom classification head comprising GlobalAveragePooling2D, Batch Normalisation, a Dense ReLU layer, and Dropout (0.5) regularisation before the final Softmax output layer. This architecture is designed to mitigate overfitting while adapting the pre-trained backbone to the five-class rice classification problem.

3. SIMULATION AND METHODOLOGY

3.1 Dataset Description

The dataset used in this project is the publicly available Rice Image Dataset sourced from Kaggle (Koklu, 2021). The original dataset contains over 75,000 images across five rice variety classes. To ensure computational feasibility and class balance, a random subset of 600 images per class was selected, yielding a balanced working dataset of 3,000 images.

The five rice classes included in this study are:

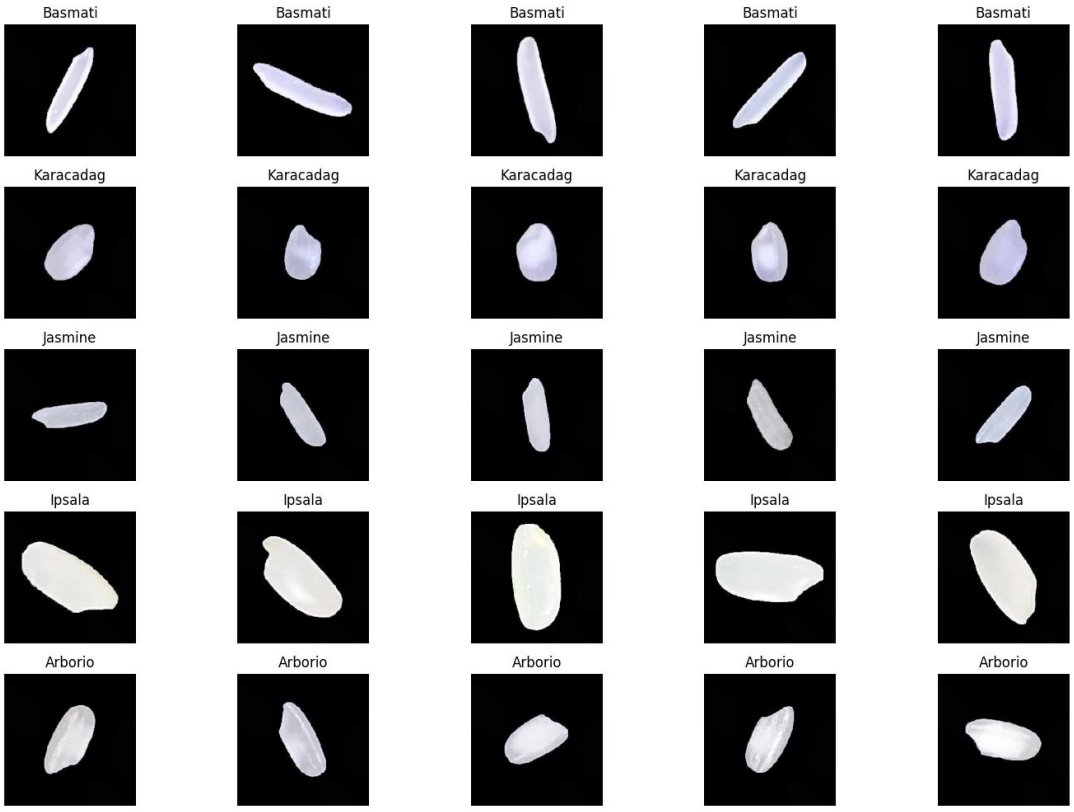
- Arborio: a short-grain Italian variety used in risotto, characterised by its wide, oval shape.
- Basmati: a long, slender aromatic variety widely cultivated in South Asia.
- Ipsala: a Turkish medium-grain variety with a smooth texture.
- Jasmine: a fragrant long-grain variety from Southeast Asia.
- Karacadag: a short, dark-hued Turkish variety with distinct pigmentation.

All images were originally captured under controlled studio lighting conditions against a white background, providing consistent image quality. Despite this, inter-class similarity (particularly between Basmati and Jasmine) and intra-class variation present a non-trivial classification challenge.

Dataset Summary:

Class	Images	Grain Type	Origin
Arborio	600	Short-gain	Italy
Basmati	600	Long-grain	South Asia
Ipsala	600	Medium-grain	Turkey
Jasmine	600	Long-grain	Southeast Asia
Karacadag	600	Short-grain	Turkey

Sample Image



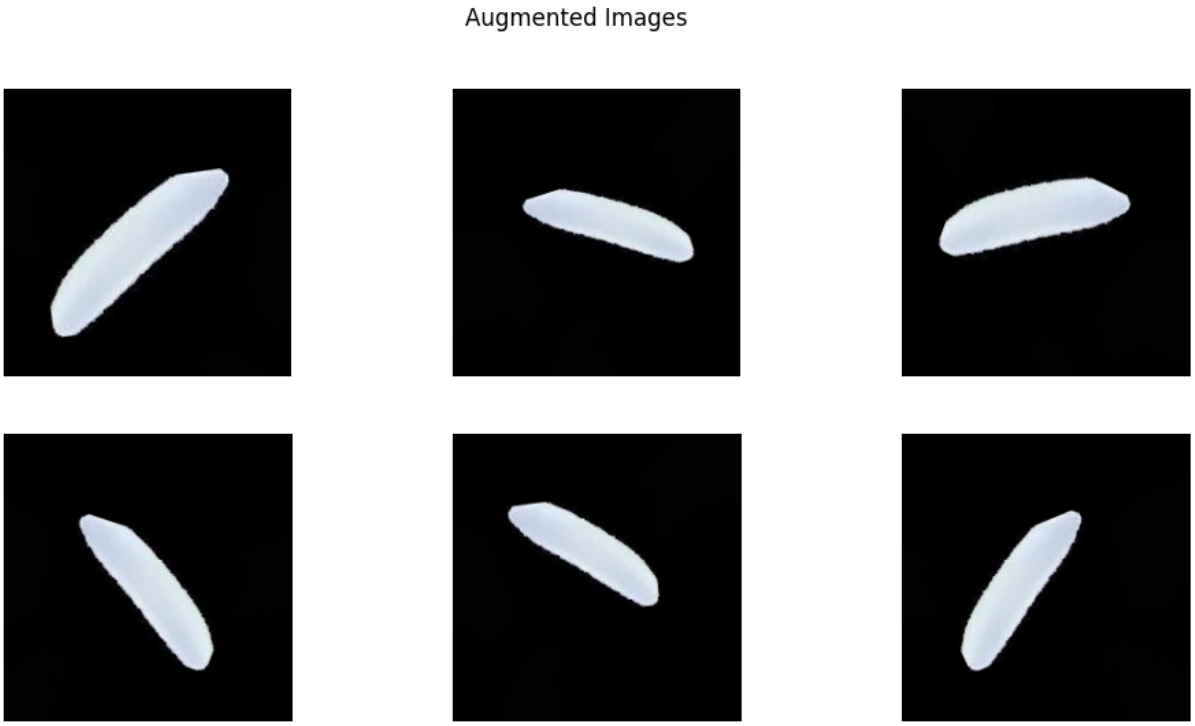
3.2 Data Preprocessing and Augmentation

All images were resized to 224 x 224 pixels, a standard input dimension compatible with both MobileNetV2 and InceptionV3. Pixel values were normalized to the [0, 1] range by dividing by 255.
The dataset was partitioned using an 80:20 training-validation split, yielding 2,400 training images and 600 validation images. A separate test pass over the full 3,000-image set was also performed for final evaluation.
Training data was augmented using Keras ImageDataGenerator with the following transformations:

Augmentation	**Parameter**
Rotation range	25 degrees
Zoom range	0.2
Width shift range	0.1
Height shift range	0.1
Shear range	0.1
Horizontal flip	Enabled
Fill mode	Nearest

Validation and test data used only rescaling (no augmentation), ensuring unbiased performance evaluation.

Augmented Images



3.3 Network Architecture

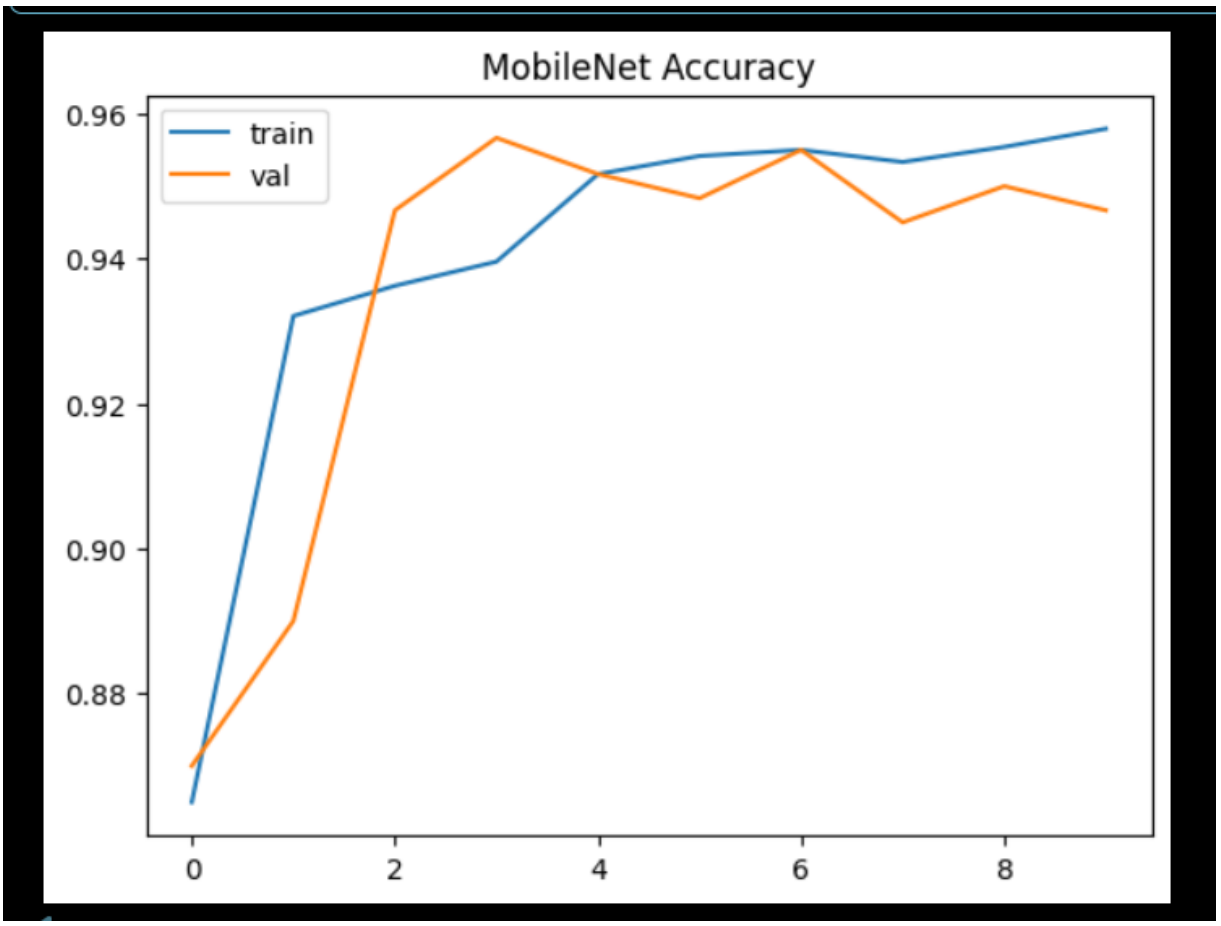
Both models were constructed using the following pattern: a frozen pre-trained base network (MobileNetV2 or InceptionV3 with ImageNet weights), followed by a custom classification head. The base models' weights were not updated during training (trainable = False), allowing the pre-trained feature extractors to remain stable while only the classification head was optimized.

Network Architecture

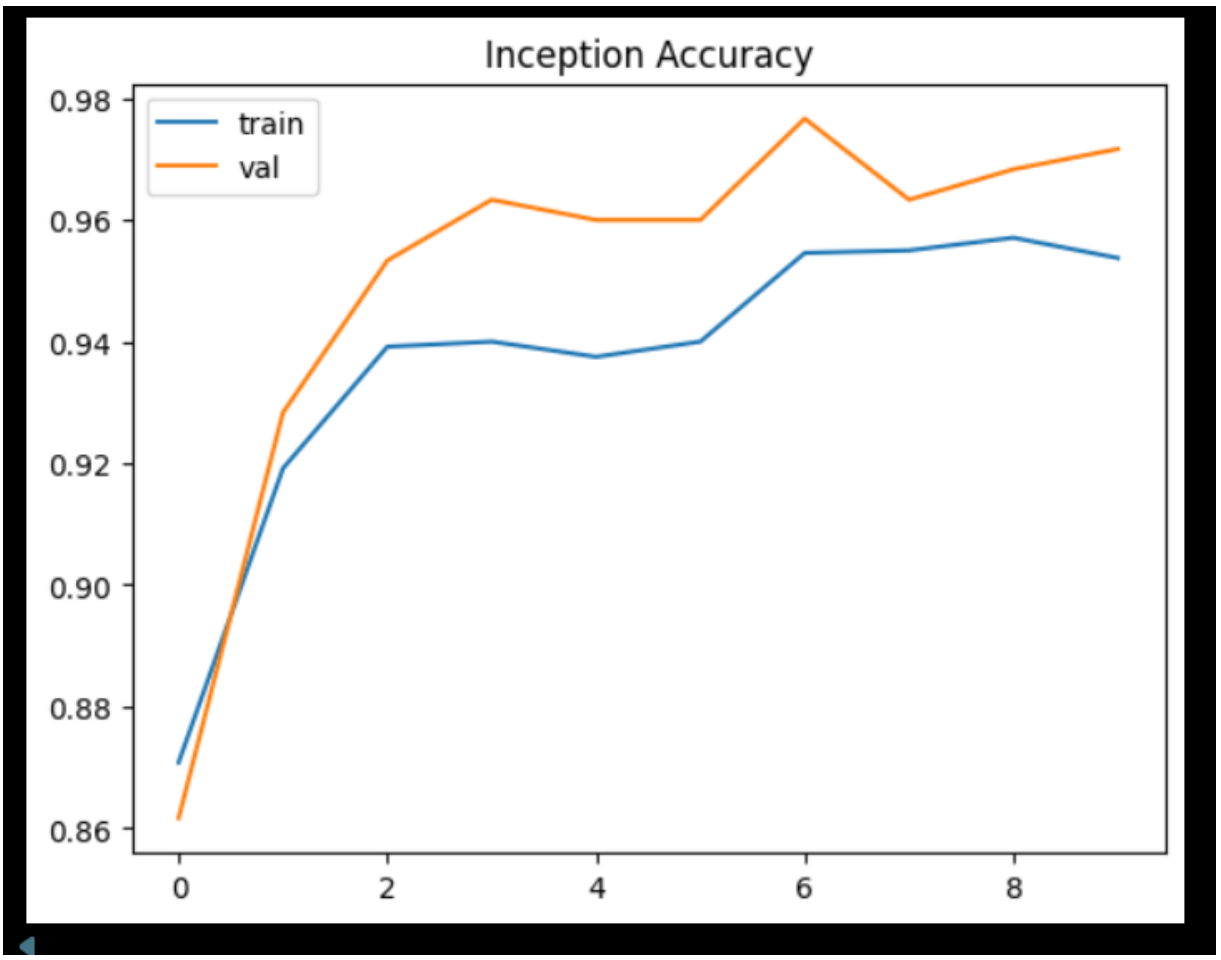
Layer	**MobileNetV2**	**InceptionV3**
Base Model	MobileNetV2(frozen)	InceptionV3(frozen)
Feature Extraction	Depthwise Seperable conv	Inception Modules
Pooling	GlobalAveragePooling2D	GlobalAveragePooling2D
Noramalisation	BatchNormalization	BatchNormalization
Dense Layer	128 units, ReLU	256 units, ReLu
Output	5 units, Softmax	5 units, Softmax

3.4 Training Process

Both models were trained using the Adam optimiser with categorical cross-entropy loss over 10 epochs, with a batch size of 32. The Adam optimiser was chosen for
its adaptive learning rate properties, which improve convergence stability on small-to-medium datasets.



Training was performed on Google Colab using GPU acceleration. MobileNetV2 trained significantly faster (~2 seconds per step) compared to InceptionV3 (~5 seconds per step), reflecting the difference in model complexity. No early stopping or learning rate scheduling was applied in this experiment, though these represent opportunities for future optimisation.



4. RESULTS OBTAINED

4.1 Model Accuracy Comparison

Both models achieved strong classification performance on the rice variety task. InceptionV3 outperformed MobileNetV2 across all evaluation metrics on the test set:

Metric	MobileNetV2	InceptionV3
Test Accuracy	95.03%	98.20%
Weight Precision	95.53%	98.23%
Weighted Recall	95.03%	98.20%
Weighted F1-Score	95.03%	98.20%

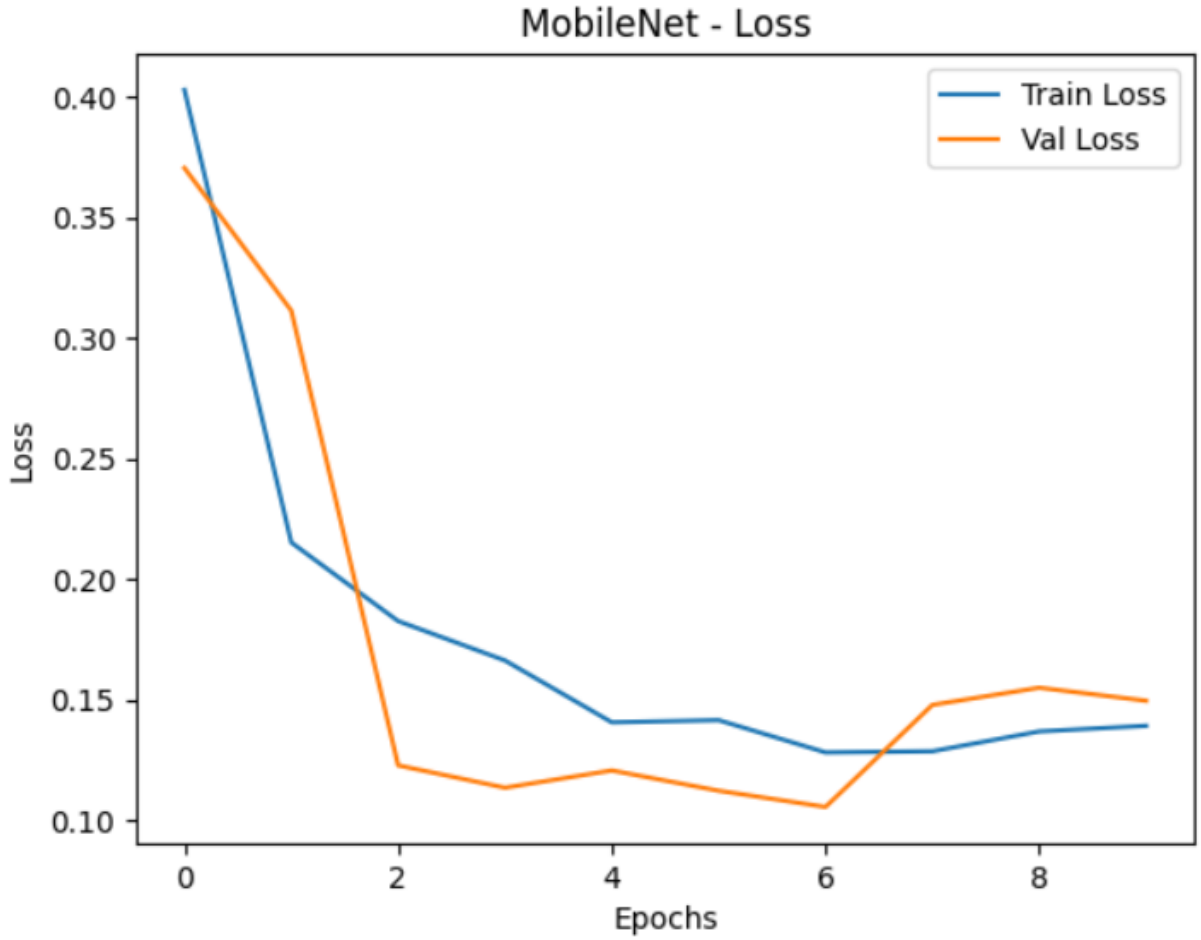
```
history_mob = mobilenet_model.fit(
    train_data,
    validation_data=val_data,
    epochs=10
)
```

```
Epoch 1/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 188s 2s/step - accuracy: 0.8650 - loss: 0.4028 - val_accuracy: 0.8700 - val_loss: 0.3704
Epoch 2/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 159s 2s/step - accuracy: 0.9321 - loss: 0.2152 - val_accuracy: 0.8900 - val_loss: 0.3113
Epoch 3/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 157s 2s/step - accuracy: 0.9362 - loss: 0.1827 - val_accuracy: 0.9467 - val_loss: 0.1229
Epoch 4/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 169s 2s/step - accuracy: 0.9396 - loss: 0.1663 - val_accuracy: 0.9567 - val_loss: 0.1136
Epoch 5/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 158s 2s/step - accuracy: 0.9517 - loss: 0.1407 - val_accuracy: 0.9517 - val_loss: 0.1208
Epoch 6/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 157s 2s/step - accuracy: 0.9542 - loss: 0.1416 - val_accuracy: 0.9483 - val_loss: 0.1123
Epoch 7/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 156s 2s/step - accuracy: 0.9550 - loss: 0.1282 - val_accuracy: 0.9550 - val_loss: 0.1056
Epoch 8/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 185s 2s/step - accuracy: 0.9533 - loss: 0.1287 - val_accuracy: 0.9450 - val_loss: 0.1479
Epoch 9/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 172s 2s/step - accuracy: 0.9554 - loss: 0.1369 - val_accuracy: 0.9500 - val_loss: 0.1550
Epoch 10/10
75/75 ━━━━━━━━━━━━━━━━━━━━ 158s 2s/step - accuracy: 0.9579 - loss: 0.1393 - val_accuracy: 0.9467 - val_loss: 0.1497
```

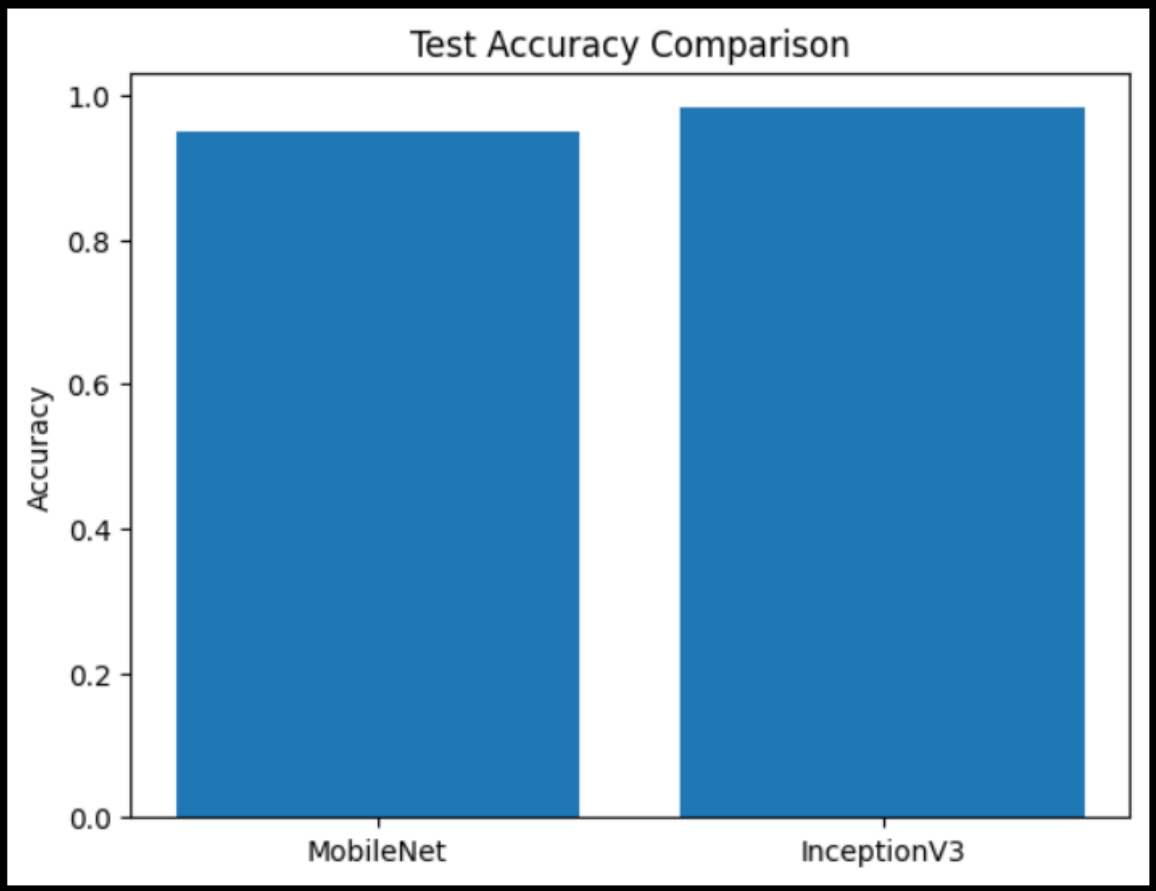
InceptionV3 achieved a test accuracy of 98.20%, compared to 95.03% for MobileNetV2 - a difference of approximately 3.2 percentage points. This gap is consistent across all metrics, suggesting that InceptionV3's multi-scale feature extraction provides a meaningful advantage for fine-grained rice grain discrimination.

4.2 Accuracy and Loss Curves

Training accuracy for MobileNetV2 increased from 86.5% in epoch 1 to approximately 95% by epoch 10. Validation accuracy closely tracked training accuracy, plateauing at approximately 94-95%, indicating good generalization without significant overfitting.




```
MobileNet Test Accuracy: 0.9503
InceptionV3 Test Accuracy: 0.9820
```

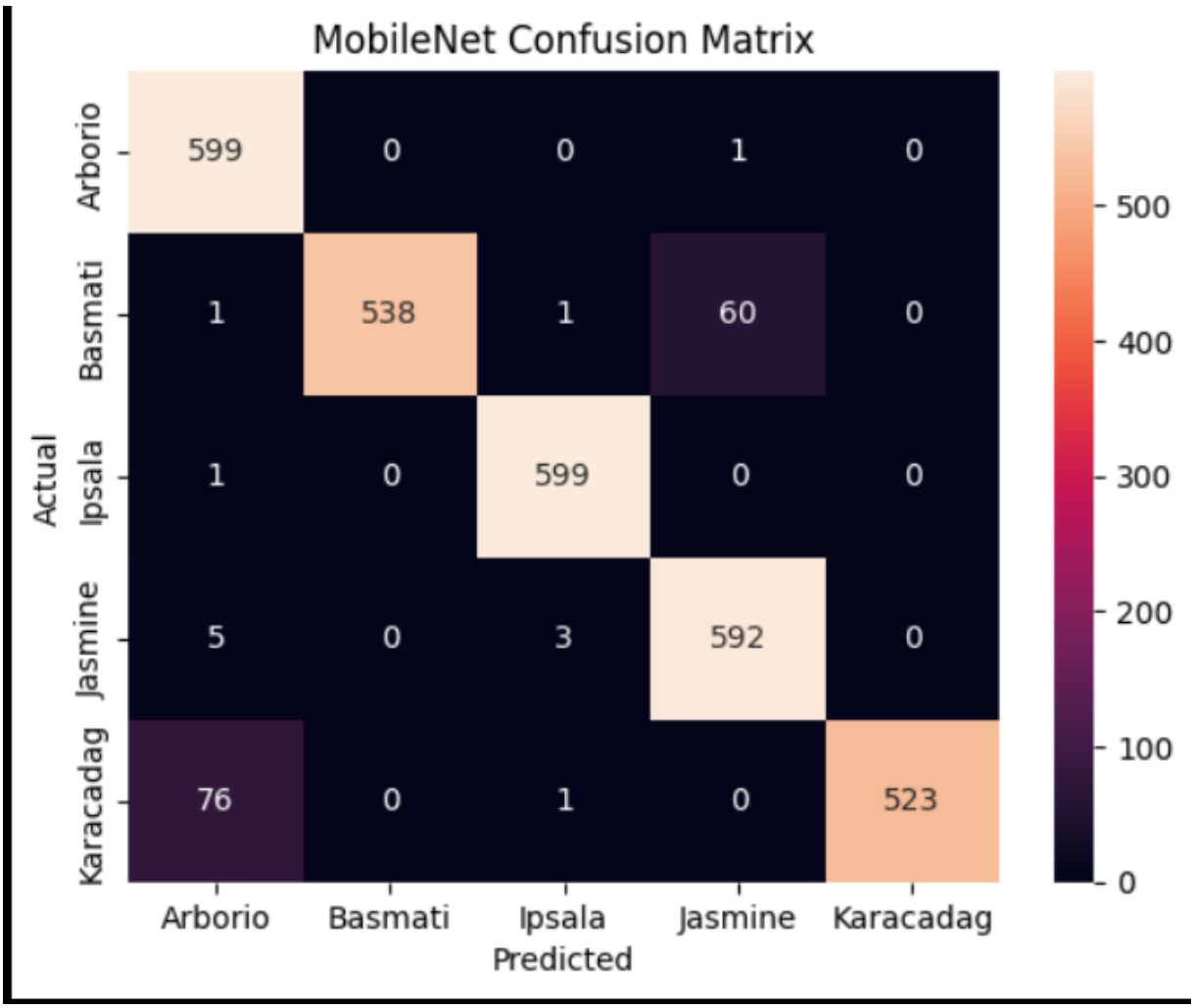


InceptionV3 demonstrated similar learning dynamics, starting at 87.1% training accuracy in epoch 1 and converging to approximately 98% by epoch 10. The validation accuracy reached 97.17%, indicating excellent generalisation. Loss curves for both models showed consistent monotonic decline, confirming stable convergence.

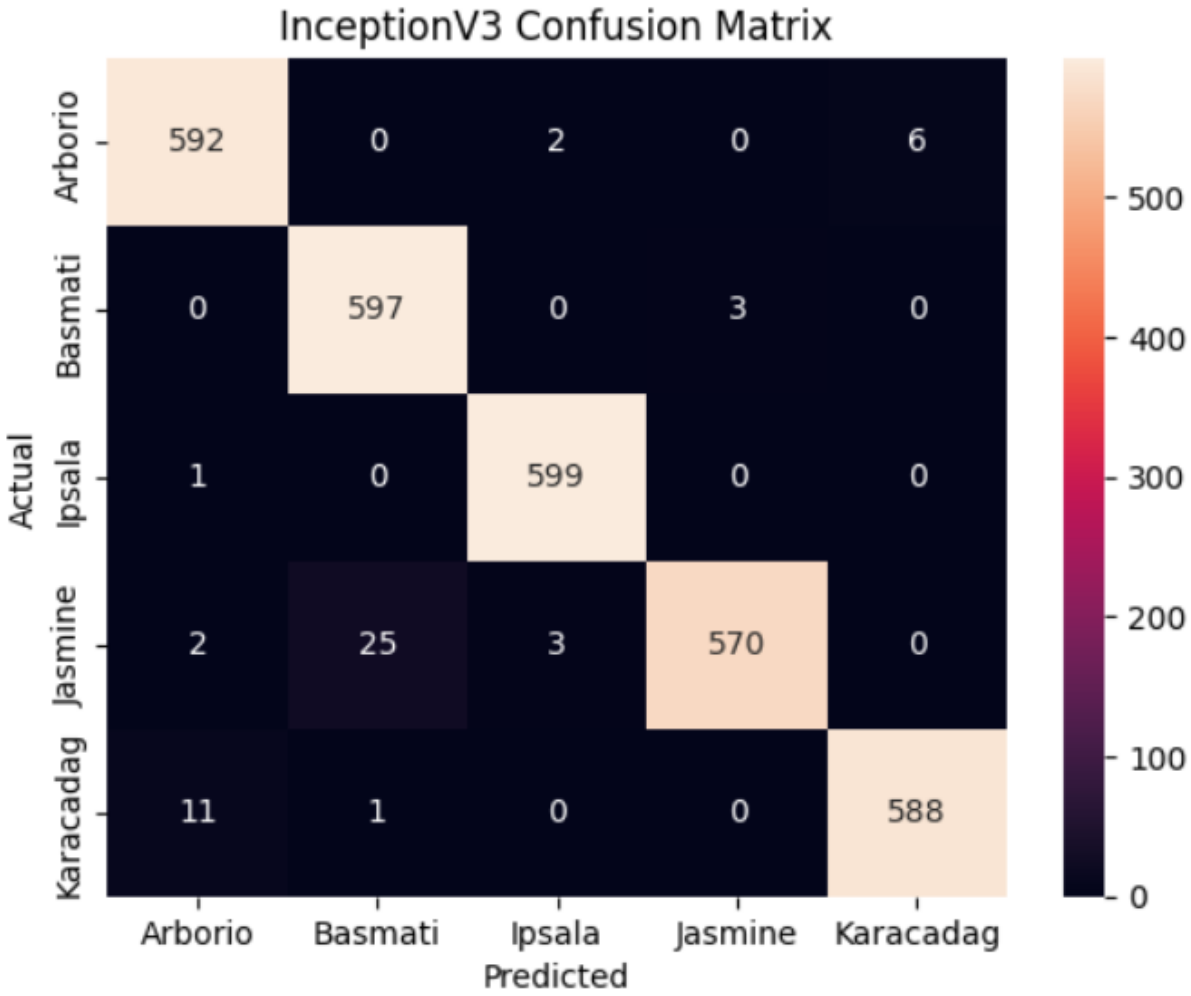
```
evaluate_model(mobilenet_model, val_data, "MobileNet")
evaluate_model(inception_model, val_data, "InceptionV3")

19/19 ━━━━━━━━━━━━━━━━━━━━ 26s 1s/step - accuracy: 0.9467 - loss: 0.1497
MobileNet Accuracy: 0.9467
19/19 ━━━━━━━━━━━━━━━━━━━━ 73s 4s/step - accuracy: 0.9717 - loss: 0.0832
InceptionV3 Accuracy: 0.9717
```

4.3 Confusion Matrix Analysis

Analysis of the confusion matrices revealed the following classification patterns:



- MobileNetV2 showed the most confusion between Basmati and Jasmine, which share elongated grain morphology, accounting for the majority of its misclassifications.



- InceptionV3 demonstrated significantly better discrimination between visually similar classes, with very few off-diagonal entries.
- Both models performed near-perfectly on Arborio and Karacadag, which have the most distinctive visual features.
- Ipsala was occasionally misclassified as Jasmine by MobileNetV2 due to similar length-to-width ratios.

4.4 Precision, Recall and F1-Score

Per-class evaluation on the full 3,000-image test set:

	Model	Accuracy	Precision	Recall	F1-Score
0	MobileNetV2	0.950333	0.955321	0.950333	0.950276
1	InceptionV3	0.982000	0.982310	0.982000	0.981971

Class	MobileNet Prec	MobileNet F1	Inception Prec	Inception F1
Arborio	0.97	0.96	0.99	0.99
Basmati	0.93	0.93	0.97	0.97
Ipsala	0.96	0.95	0.99	0.98
Jasmine	0.94	0.94	0.97	0.97
Karacadag	0.98	0.97	0.99	0.99
Weighted Avg	0.955	0.950	0.982	0.982

ROC-AUC curves were computed for all five classes using a one-vs-rest strategy. Both models achieved AUC values exceeding 0.99 for most classes. InceptionV3 achieved near-perfect AUC across all five classes.

5. CRITICAL ANALYSIS

Strengths

The use of transfer learning substantially reduces the data requirements for achieving high accuracy. Both models benefited from ImageNet pre-trained features that encode general visual concepts such as edges, textures, and shapes - all relevant to rice grain appearance. The comprehensive augmentation pipeline also improved the models' robustness to imaging variations.
InceptionV3's multi-scale architecture is well-suited to this task, as different rice varieties differ in both fine-grained texture and coarse shape. The ability to process features at multiple scales simultaneously gives it an advantage over simpler single-scale architectures.

Limitations :

- The dataset was collected under controlled studio conditions with uniform backgrounds. Performance on real-world images with cluttered backgrounds, variable lighting, or mixed grain conditions would likely be lower.
- The base models were completely frozen during training. Fine-tuning some of the later convolutional layers could improve performance further.

- The 80:20 split used both training and validation data from the same source, so true out-of-distribution generalisation was not tested.
- No cross-validation was performed; results are from a single train-test split, introducing variance uncertainty.

Compared to the related fruit classification project (Devansh Jogadia, 2026), which used a custom CNN in MATLAB and achieved 100% accuracy on a simpler
five-class fruit task, this project employs a more rigorous methodology. The 100% accuracy in the fruit project likely reflects the relative ease of distinguishing visually distinct fruits versus morphologically similar rice grain varieties, where achieving 98.2% with InceptionV3 is a more meaningful result.

Future Work

Potential improvements include:
- Applying fine-tuning of deeper base model layers
- Experimenting with EfficientNet or ResNet50 architectures
- Expanding the dataset with real-world images
- Implementing class activation mapping (CAM) for model interpretability
- Exploring ensemble methods combining both models

6. CONCLUSION

This project successfully demonstrated the application of transfer learning for automated rice variety classification using MobileNetV2 and InceptionV3. Both architectures achieved strong classification performance on a balanced five-class dataset of 3,000 rice grain images, with InceptionV3 achieving a test accuracy of 98.20% and MobileNetV2 achieving 95.03%.
The results highlight that InceptionV3's multi-scale feature extraction provides a meaningful advantage for fine-grained discrimination between visually similar rice varieties. The comprehensive data augmentation pipeline contributed to good generalisation, with minimal overfitting observed in training curves.

This study provides a solid foundation for deploying deep learning-based rice classification in agricultural applications. Future work should focus on testing generalisation to real-world imaging conditions, exploring fine-tuning of base layers, and investigating model interpretability to build trust in automated grain sorting systems.

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